Lie algebras and their representations

Based on a lecture series by Ian Grojnowski.

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This is a course about symmetries controlled by the *linear algebraic groups*

$$\mathsf{GL}_n(\mathbb{C}) = \{A \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \det A \neq 0\}, \qquad \mathsf{O}_n(\mathbb{C}) = \{A \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid A^{\mathsf{T}} \cdot A = \mathrm{id}\},$$
$$\mathsf{SL}_n(\mathbb{C}) = \{A \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \det A = 1\}, \qquad \mathsf{SP}_n(\mathbb{C}) = \{A \in \mathrm{Mat}_{n\times n}(\mathbb{C}) \mid \langle x, y\rangle = \langle Ax, Ay\rangle \ \forall x, y \in \mathbb{C}^n\}$$

(here the *Symplectic group* $\mathsf{SP}_n(\mathbb{C})$ is defined when n is even, and the given definition depends on the choice of a non-degenerate skew-symmetric bilinear form $\langle -, -\rangle \colon \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$), and five others — which happen to be groups equipped with additional structures of a smooth manifold and an algebraic variety. These structures mean we can use topological and analytic methods to study the groups; on the other hand the structures also simplify much of the study of these groups to simple linear algebra. This reduction is accomplished by associating the group to its tangent space at identity, which comes naturally equipped with the structure of a *Lie algebra* that shadows the group operation.

This course is about Lie algebras and their representation theory over a field of characteristic zero.

§ 0.1 About. This course on Lie algebras is a part of Part III of the Mathematics tripos at the University of Cambridge. The lectures were delivered in-person, and were transcribed live. The notes can be found online at `https://pas201.user.srcf.net/documents/2026-lie-algebras.pdf`.

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§ 1 Getting started

Lecture 1, 08/10/26

The Lie algebra associated to an algebraic group G is the tangent space $\mathfrak{g} = \mathsf{T}_{\mathrm{id}}\mathsf{G}$ at the identity element. The group structure on G induces the structure of a *Lie algebra* on $\mathfrak{g}$. We unpack this with an ad hoc definition first.

Definition 1.1 (Ad hoc). A complex *linear algebraic group* G is a subvariety of the affine space $\mathrm{Mat}_{n\times n}(\mathbb{C})$ of $n \times n$ matrices, defined as the zero locus of some polynomials in matrix coefficients, equipped with an associative multiplication map $\mathsf{G} \times \mathsf{G} \to \mathsf{G}$ and an inverse map $\mathsf{G} \to \mathsf{G}$ defined by polynomials in matrix coefficients, such that the usual axioms of a group are satisfied.

This is a terrible definition, because it is not *intrinsic*. It relies on choosing an embedding of the group into a coordinate system. There *is* an intrinsic definition, under which a linear algebraic group is simply a group-object in the category of affine varieties.

Remark 1.2. It is not obvious that every such intrinsically defined object can be embedded into some affine space $\mathrm{Mat}_{n\times n}(\mathbb{C})$, but it follows from noting that G would then act on the space of functions on G faithfully, and by Hilbert's basis theorem there is a finite dimensional faithful sub-representation therein.

Remark 1.3. Each linear algebraic group has a maximal compact subgroup (unique up to conjugacy), and the linear algebraic group is diffeomorphic to the tangent bundle over this compact subgroup. Note that the tangent bundle of any Lie group is trivial.

For instance the maximal compact subgroup of $\mathsf{SL}_n(\mathbb{C})$ is the special unitary group $\mathsf{SU}_n(\mathbb{C})$, diffeomorphic to an n-sphere. Likewise the maximal compact subgroup inside $\mathsf{SO}_n(\mathbb{C})$ is $\mathsf{SO}_n(\mathbb{R})$.

What is a tangent space? Let us begin with an example, by calculating the tangent space to $\mathsf{SL}_2(\mathbb{C})$ at $\mathrm{id} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. An element close to id can be expressed as

$$P = \mathrm{id} + \varepsilon \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) + \varepsilon^2(\cdots),$$

so for this to lie in $\mathsf{SL}_2(\mathbb{C})$ we must have $\det(P) = 1$, which expands to

$$1 + \varepsilon(a + d) + \varepsilon^2(\cdots) = 1.$$

To compute the tangent space we wish for ε to be "very small", so that $\varepsilon^2 \approx 0$ and the tangent space is this first-order neighbourhood

$$\mathfrak{sl}_2(\mathbb{C}) = \{\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \mid a + d = 0\}.$$

This construction can be made rigorous by working over the ring $\mathbb{E} = \mathbb{C}[\varepsilon]/(\varepsilon^2)$. Recall that if G is a complex linear algebraic group and R is a $\mathbb{C}$-algebra, we define the *set of* R*-points of* G as

$$G(R) = \{P \in \mathrm{Mat}_{n\times n}(R) \mid P \text{ satisfies the polynomial equations defining } G \subset \mathrm{Mat}_{n\times n}(\mathbb{C})\}$$

There is a map $\mathbb{E} \to \mathbb{C}$ sending $\varepsilon \mapsto 0$, this in turn induces a map $\pi\colon G(\mathbb{E}) \to G(\mathbb{C})$. The tangent space $\mathfrak{g} = \mathsf{T}_{\mathrm{id}}G$ is then defined as the pre-image $\pi^{-1}(\mathrm{id})$.

Exercise 1.4. Show that $\mathsf{GL}_n(\mathbb{C})$ is a linear algebraic group (*hint: the condition* $\det P \neq 0$ *can be written as the vanishing of a polynomial by embedding* $\mathsf{GL}_n(\mathbb{C})$ *into* $\mathrm{Mat}_{(n+1)\times(n+1)}(\mathbb{C})$). Use this to show that the associated Lie algebra is $\mathfrak{gl}_n(\mathbb{C}) = \mathrm{Mat}_{n\times n}(\mathbb{C})$.

Exercise 1.5. For an $n \times n$ matrix X, show that $\det(\mathrm{id} + \varepsilon X) = 1 + \varepsilon \cdot \mathrm{tr}(X)$. Conclude that the Lie algebra $\mathfrak{sl}_n(\mathbb{C})$ associated to $\mathsf{SL}_n(\mathbb{C})$ is the space of traceless $n \times n$ matrices.

Example 1.6. For the orthogonal group over $\mathbb{C}$, the associated Lie algebra is the space of antisymmetric matrices. Note that all such matrices are traceless over any field of characteristic $\neq 2$, thus $\mathsf{SO}(n)$ has the same Lie algebra. We therefore denote it as $\mathfrak{so}_n(\mathbb{C})$.

In the above example we have learnt that different groups can have the same Lie algebra. This shouldn't be surprising — for instance, all finite groups are linear algebraic groups, and their Lie algebra is the zero algebra $\mathbf{0}$. In a similar fashion the orthogonal group is disconnected, with two connected components diffeomorphic to each other, one of which is $\mathsf{SO}_n(\mathbb{C})$.

One thing we will learn in due course is that a Lie group, its maximal compact subgroup, and its Lie algebra all have coincident finite dimensional representations.

The Lie bracket. What structure does $\mathfrak{g} = \mathsf{T}_{\mathrm{id}}G$ have that is inherited from the group structure on G? It is not multiplication.

What we do instead is to look at the commutator map $G \times G \to G$ sending $(p, q) \mapsto p^{-1}q^{-1}pq$. This sends $(\mathrm{id}, \mathrm{id}) \mapsto \mathrm{id}$, thus the derivative of the commutator gives a linear map

$$\mathfrak{g} \times \mathfrak{g} \simeq \mathsf{T}_{(\mathrm{id},\mathrm{id})}(G \times G) \longrightarrow \mathsf{T}_{\mathrm{id}}G \simeq \mathfrak{g}.$$

This is simply the zero map, and is not quite what we want — we want a *bilinear* map $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$, so we differentiate the first and the second variable separately.

To explain the construction, suppose G is $\mathsf{GL}_n(\mathbb{C})$ or another matrix group. Let $P = \mathrm{id} + X\varepsilon$, $Q = \mathrm{id} + Y\delta$ where ε, δ are small and satisfy $\varepsilon^2 = \delta^2 = 0$ but $\varepsilon\delta = \delta\varepsilon \neq 0$ (i.e. we work over the ring $\Bbbk[\varepsilon, \delta]/(\varepsilon^2, \delta^2)$). We then see that

$$P^{-1}Q^{-1}PQ = \mathrm{id} + \varepsilon\delta(XY - YX),$$

so it makes sense to introduce the map $[\,,\,]\colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ given by $[X, Y] = XY - YX$. This is the *Lie bracket.*

Exercise 1.7. The above calculation is an extrinsic proof of the fact that the Lie bracket $[\,,\,]\colon \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$ is *antisymmetric.* To get an intrinsic proof that doesn't rely on any embedding into matrix groups, use the fact that $(pqp^{-1}q^{-1})^{-1} = qpq^{-1}p^{-1}$ for any group elements p, q.

The next property of the Lie bracket is the *Jacobi identity*:

$$[[x,y],z]+[[y,z],x]+[[z,x],y]=0 \qquad \text{for all } x,y,z\in\mathfrak{g}.$$

We will deduce this over multiple exercises, seeing three reasons why it is true. Next time we will look at a fourth and a fifth explanation.

Exercise 1.8. For $G\subseteq \mathrm{GL}(n,\mathbb{C})$, show by direct computation that the Jacobi identity holds.

Exercise 1.9. Marshall Hall discovered the following identity in the 1930s: writing $p^q = q^{-1}pq$ and $(p,q)=p^{-1}q^{-1}pq$, Hall showed that in any group G with elements p,q,r we have

$$((p,q^{-1}),r)^q\cdot((q,r^{-1}),p)^r\cdot((r,p^{-1}),q)^p=\mathrm{id}.$$

Differentiate this to deduce the Jacobi identity.

Exercise 1.10. Deduce the Jacobi identity from associativity of the group law on G.

Lie algebras. The two properties above — antisymmetry and the Jacobi identity — define an abstract Lie algebra. Fix a field $\Bbbk$ of characteristic $\neq 2,3$.

Definition 1.11. Let $\mathfrak{g}$ be a $\Bbbk$-vector space. A *Lie bracket* on $\mathfrak{g}$ is an antisymmetric bilinear form $[\,,\,]\colon \mathfrak{g}\times\mathfrak{g}\to\mathfrak{g}$ that satisfies the Jacobi identity. We say $\mathfrak{g}$ equipped with a Lie bracket is a *Lie algebra.*

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Exercise 1.12. Here is another way to define a Lie bracket on a $\Bbbk$-vector space $\mathfrak{g}$ — it is given by an element $\mu\in\bigwedge^2(\mathfrak{g}^\vee)\otimes\mathfrak{g}$ (i.e. a $\mathfrak{g}$-valued 2-form on $\mathfrak{g}^\vee$) which lies in the kernel of the induced exterior derivative $d_\mu\colon \bigwedge^\bullet(\mathfrak{g}^\vee)\otimes\mathfrak{g}\to\bigwedge^{\bullet+1}(\mathfrak{g}^\vee)\otimes\mathfrak{g}$. Recall that this derivative is given by $d_\mu(1)=0$, $d_\mu\omega=\omega\circ\mu$ for $\omega\in\mathfrak{g}^\vee\otimes\mathfrak{g}$, extended by the graded Leibniz rule to the rest of $\bigwedge^\bullet(\mathfrak{g}^\vee)\otimes\mathfrak{g}$. Check the equivalence of the two definitions.

Example 1.13. If V is a finite dimensional $\Bbbk$-vector space, then $\mathfrak{gl}(V)=\mathrm{End}_\Bbbk(V)$ is a Lie algebra, isomorphic to $\mathfrak{gl}_n(\Bbbk)$. We omit the field from the notation unless necessary, thus simply writing $\mathfrak{gl}_n$.

Other examples include the Lie algebras

$$\begin{aligned}
\mathfrak{sl}_n &= \{X\in\mathfrak{gl}_n \mid \mathrm{tr}(X)=0\},\\
\mathfrak{so}_n &= \{X\in\mathfrak{gl}_n \mid X+X^{\mathsf{T}}=0\},\\
\mathfrak{sp}_n &= \{X\in\mathfrak{gl}_n \mid JX^{\mathsf{T}}J^{-1}+X=0\} \qquad (n \text{ even, } J \text{ the standard } n\times n \text{ symplectic matrix}),\\
\mathfrak{b}_n &= \{X\in\mathfrak{gl}_n \mid X \text{ upper triangular}\},\\
\mathfrak{n}_n &= \{X\in\mathfrak{gl}_n \mid X \text{ strictly upper triangular}\}.
\end{aligned}$$

Exercise 1.14. Check directly that each of the above, equipped with the commutator bracket, is a Lie algebra. Which algebraic groups are they the Lie algebras of?

Exercise 1.15. Show that $\{\left(\begin{smallmatrix} * & * \\ & * \end{smallmatrix}\right)\}$ is not a Lie subalgebra of $\mathfrak{gl}_2$.

Example 1.16. If V is a $\Bbbk$-vector space, then the zero form $0\in\bigwedge^2(V^\vee)\otimes V$ makes V into a trivial Lie algebra. Such Lie algebras are termed *Abelian.*

Exercise 1.17. Classify one, two, and three-dimensional Lie algebras over $\mathbb{C}$.

§ 2 Representations of Lie algebras

Definition 2.1. A *representation* of a Lie algebra $\mathfrak{g}$ on a $\Bbbk$-vector space V is a homomorphism of Lie algebras $\rho\colon\mathfrak{g}\to\mathfrak{gl}(V)$. In this case we say "$\mathfrak{g}$ acts on V".

For example, each Lie algebra $\mathfrak{g}$ acts on itself — given $x\in\mathfrak{g}$, define the map $\mathrm{ad}_x\colon\mathfrak{g}\to\mathfrak{g}$ as $y\mapsto[x,y]$.

Lemma 2.2. *The map* $\mathfrak{g} \to \mathfrak{gl}(\mathfrak{g})$ *given by* $x \mapsto \mathrm{ad}_x$ *is a representation of* $\mathfrak{g}$. *We call this the* adjoint representation.

Proof. We must show $\mathrm{ad}_{[x,y]} = \mathrm{ad}_x \circ \mathrm{ad}_y - \mathrm{ad}_y \circ \mathrm{ad}_x$. But $\mathrm{ad}_{[x,y]}(z) = [[x,y],z]$, while the right-hand side sends z to $[x,[y,z]] - [y,[x,z]]$. By antisymmetry, the required result is equivalent to the Jacobi identity. □

Definition 2.3. The *center* of $\mathfrak{g}$ is the kernel of the adjoint action $\ker(\mathrm{ad}\colon \mathfrak{g} \to \mathfrak{gl}(\mathfrak{g}))$

Thus a Lie algebra with trivial center acts faithfully on itself.

Remark 2.4. In fact *every* finite dimensional Lie algebra has a faithful finite dimensional representation. This is a non-trivial theorem of Ado, and the construction of the representation is non-obvious (unlike the corresponding statement for algebraic groups, where you can simply consider the space of functions on the group). We will look at the proof if time permits.

Example 2.5. The Lie algebra $\mathfrak{sl}_2 = \{\left(\begin{smallmatrix} a & b \\ c & -a \end{smallmatrix}\right) \mid a,b,c \in \Bbbk\}$ has a basis given by the three elements

$$e = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \quad h = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right), \quad f = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$$

which satisfy the relations

$$[h,e] = 2e, \quad [h,f] = -2f, \quad [e,f] = h.$$

Hence a representation of $\mathfrak{sl}_2$ on $\Bbbk^n$ is given by a triple $E,F,H \in \mathrm{Mat}_{n\times n}(\Bbbk)$ such that $EF - FE = H$, $HF - FH = -2F$, and $HE - EH = 2E$.

Exercise 2.6. If G is an algebraic group, an (algebraic) representation of G on a $\Bbbk$-vector space V is a group homomorphism $\rho\colon G \to \mathrm{GL}(V)$ defined by polynomial equations in matrix coefficients. For the ring $\mathbb{E} = \Bbbk[\varepsilon]/(\varepsilon^2)$, we thus have a map of the groups of $\mathbb{E}$-points $G(\mathbb{E}) \to \mathrm{GL}(V)(\mathbb{E})$ satisfying $\rho(\mathrm{id}) = \mathrm{id}$. It follows that $\rho(\mathrm{id} + X\varepsilon) = \mathrm{id} + d\rho(X)\cdot\varepsilon$ for some matrix $d\rho(X)$. Check that this assignment $d\rho\colon \mathsf{T}_{\mathrm{id}}G \to \mathfrak{gl}(V)$ is a Lie algebra homomorphism.

Example 2.7. Take $G = \mathrm{SL}_2(\Bbbk)$, and $L(n) = \mathrm{Sym}^n(\Bbbk^{\oplus 2})$ (equivalently, the degree two homogeneous piece of $\Bbbk[x,y]$). As $\mathrm{GL}_2(\Bbbk)$ acts on $L(n)$, so does $\mathrm{SL}_2(\Bbbk)$. Call this representation $\rho_n\colon \mathrm{SL}(2,\Bbbk) \to \mathrm{GL}(L(n))$. For example, ρ_0 is the trivial representation, ρ_1 is the standard representation, and we have

$$\rho_2\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad+bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}.$$

Differentiating, we get representations $\mathfrak{sl}_n \to \mathfrak{gl}(L(n))$ which we simply refer to as $L(n)$.

Exercise 2.8. Writing $(x^i y^j)_{i+j=n}$ for the standard basis of $L(n)$, show that we have

$$e \cdot x^i y^j = j x^{i+1} y^{j-1}, \quad f \cdot x^i y^j = i x^{i-1} y^{j+1}, \quad h \cdot x^i y^j = (i-j) x^i y^j,$$

$$\text{or equivalently,} \qquad e = x\frac{\partial}{\partial y}, \quad f = y\frac{\partial}{\partial x}, \quad h = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

Assuming $\mathrm{char}(\Bbbk) = 0$, show that each $L(n)$ is an irreducible representation.

Remark 2.9. How to think about these representations $L(n)$? The space $L(n)$ is $H^0(\mathbb{P}^1, \mathcal{O}(n))$. This is significant because $\mathbb{P}^1$ is a homogeneous space for the group SL_2, and $\mathfrak{sl}_2$ is precisely the Lie algebra of holomorphic vector fields on $\mathbb{P}^1$. A second approach to defining Lie algebras and their representations is thus to consider homogeneous spaces for other algebraic groups.

Lecture 3, 13/10/26

Example 2.10. Take the group $G = \mathbb{C}^\times$ with its lie algebra $\mathfrak{g} \simeq \mathbb{C}$. A representation of $\mathfrak{g}$ on a vector space V is an element $A \in \mathrm{End}(V)$, and two such elements A, A' define isomorphic representations if and only they are conjugate. Thus the isoclasses of $\mathfrak{g}$-representations are in bijection with Jordan normal forms of matrices. Likewise a subspace $W \subset V$ is a subrepresentation if and only if it is preserved by A, and since any matrix always has an eigenvector, the irreducible (i.e. simple) $\mathfrak{g}$-representations are all one-dimensional. In contrast, irreducible representations of G are all one-dimensional, but of the form $\rho_n\colon z \mapsto z^n$, which have derivative $d\rho_n = n$. Thus out of the $\mathbb{C}$-many irreducible representations of $\mathfrak{g}$, only $\mathbb{Z}$-many come from the Lie group G.

The situation for $\mathfrak{sl}_2$ is drastically different.

Theorem 2.11 (Lie). *If G is a simply connected (almost) simple algebraic group over $\mathbb{C}$ (or a compact Lie group), with Lie algebra $\mathfrak{g}$, then the natural map $\mathrm{Rep}(G) \to \mathrm{Rep}(\mathfrak{g})$ is an equivalence of categories.*

Here, *almost simple* means that the group has no nontrivial connected normal subgroup. For example, $\mathsf{SL}_n(\mathbb{C})$ has centre $\{\pm\mathrm{id}\}$. This is an important constraint because quotienting by disconnected discrete normal subgroups does not affect the Lie algebra. The quotient group $\mathsf{SL}_n(\mathbb{C})/\{\pm\mathrm{id}\} \simeq \mathsf{PGL}_n(\mathbb{C})$ has the same Lie algebra as $\mathfrak{sl}_2$. This is consistent with the fact that every PGL_n-representation is naturally a SL_n-representation (but this doesn't account for all SL_n-representations).

Remark 2.12. Not all finite dimensional Lie algebras come from Lie groups. In characteristic zero, there is a five-dimensional example somewhere in Bourbaki. In characteristic $p > 0$ it is simpler to see — noting that the $p \times p$ identity matrix is traceless, the quotient $\mathfrak{sl}_p/(\Bbbk \cdot \mathrm{id})$ can be shown to not arise from a Lie group.

§ 2.1 Classifying representations of $\mathfrak{sl}_2(\mathbb{C})$. Let e, f, h be the standard basis of $\mathfrak{sl}_2(\mathbb{C})$, with commutation relations $[e, f] = h$, $[h, e] = 2e$, $[h, f] = -2f$.

Theorem 2.13. *Every finite dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$ is a direct sum of irreducible representations. For all $n \geqslant 0$, there exists a unique irreducible representation of $\mathfrak{sl}_2$ that is $(n+1)$-dimensional.*

We will build up to the proof, by first making some definitions. Thus let V be a finite dimensional representation of $\mathfrak{sl}_2$.

Definition 2.14. For any complex number λ, the *λ-weight space* for V (denoted V_λ) is defined as the λ-eigenspace for the operator $h\colon V \to V$.

Suppose $v \in V_\lambda$. Then the commutation relations imply $ev \in V_{\lambda+2}$ and $fv \in V_{\lambda-2}$. Thus the actions of e and f move the vectors 'up and down' the h-eigenspaces.

Definition 2.15. If a non-zero vector v lies in $V_\lambda \cap \ker(e)$, i.e. satisfies $hv = \lambda v$ and $ev = 0$, then we say that v is a *highest weight vector of weight* λ.

Lemma 2.16. *Let $v \in V_\lambda$ be a highest weight vector of weight λ. Let $W = \langle v, fv, f^2v, \ldots \rangle \subseteq V$. Then W is a subrepresentation of V.*

Proof. We need to show $fW \subseteq W$, $eW \subseteq W$, and $hW \subseteq W$. The first and third statements are immediate. To show $eW \subseteq W$, show by induction that we have $ef^n v = n(\lambda - n + 1)f^{n-1}v$, from which the conclusion will follow. □

Lemma 2.17. *Let $v \in V_\lambda$ be a highest weight vector of weight λ. Then $\lambda \in \mathbb{Z}_{\geqslant 0}$ is a non-negative integer.*

Proof. All $f^k v$ ($k \geqslant 0$) lie in distinct eigenspaces of h, so that they are linearly independent. Since V is finite dimensional, there is a k with $f^k v \neq 0$ but $f^{k+1}v = 0$. But then we have

$$0 = ef^{k+1}v = (k+1)(\lambda - k)f^k v,$$

and since $(k+1)(f^k v) \neq 0$, we must have $\lambda = k$. □

Lemma 2.18. *Every finite dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$ has a highest weight vector.*

Proof. Since V is a $\mathbb{C}$-vector space, the operator h has an eigenvector v. Then $v, ev, e^2v, \ldots$ are all vectors which, if non-zero, are linearly independent since they lie in distinct eigenspaces of h. But V is finite dimensional, so there is a k with $e^k v \neq 0$, $e^{k+1}v = 0$. Then $e^k v$ is a highest weight vector. □

Corollary 2.19. *If V is an irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ of dimension $n+1$ ($n \geqslant 0$), then V has a basis $v_0, v_1, \ldots, v_n$ with $hv_i = (n-2i)v_i$, $fv_i = v_{i+1}$, and $ev_i = i(n-i+1)v_{i-1}$. In particular there is a unique $(n+1)$-dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$.*

Exercise 2.20. Write an explicit isomorphism between $L(n)$ and the irreducible representation of $\mathfrak{sl}_2(\mathbb{C})$ constructed above.

We move on to proving *complete reducibility* in $\mathrm{Rep}(\mathfrak{sl}_2\mathbb{C})$, i.e. showing that every indecomposable representation is irreducible. When proving analogous results for finite groups or compact Lie groups, the trick is to take some average over all group elements. We don't have such tools here.

Definition 2.21. Let V be a representation of $\mathfrak{sl}_2(\mathbb{C})$. The associated *Casimir operator*, denoted $\Omega\colon V \to V$, is given by $ef + fe + \frac{1}{2}h^2$.

Lemma 2.22. *The operator Ω is central, i.e. it commutes with e, f and h.*